Abstract

Uranium is a toxic and pervasive geogenic contaminant often associated with soil organic matter. Its abundance and speciation in organic-rich permafrost soils are unknown, thereby limiting our ability to assess risks associated with uranium mobilization during permafrost thaw. In this study, we assessed uranium speciation in permafrost soil and porewater liberated during thaw using active-layer and permafrost samples from a study area in Yukon, Canada where elevated uranium concentrations occur in bedrock and groundwater. Permafrost contained 1.1–28 wt. % organic carbon and elevated uranium (range 7.6–1,040 µg g^{-1}, median 25 µg g^{-1}) relative to local bedrock. The highest soil uranium concentrations were encountered in catchments hosting uranium-enriched bedrock, and correlated positively with soil organic carbon. X-ray absorption spectroscopy, micro-X-ray fluorescence, and electron microscopy analyses revealed that uranium predominantly occurs as uranium(VI) associated with soil organic matter. Extended X-ray absorption fine structure (EXAFS) analyses suggested the presence of uranium(VI) coordinated with carbon, consistent with bidentate-mononuclear uranyl complexation on carboxyl groups. Permafrost thaw produced circumneutral pH porewater (pH 6.2–7.5) with elevated dissolved uranium (0.5–203 µg L^{-1}). Geochemical modeling indicated that calcium-uranyl-carbonate complexes dominated dissolved uranium speciation. This study indicates that permafrost soil can mobilize uranium upon thaw, and that uranium fate is linked to dynamic biogeochemical reactions involving organic carbon and groundwater chemistry.